VectorQuay
INDIGO SIGNAL

PRODUCT REQUIREMENTS DOCUMENT

Phase 1 PRD

Repository, security, documentation, and GUI foundation for the first implementation phase.

VERSION	STATUS
v1.0	**Approved**
OWNER	THEME
Brad Malia	**Indigo Signal**

Executive Summary

Phase 1 is the first implementation phase. It does not build trading behavior yet. It establishes the local product foundation for VectorQuay: repository structure, secure configuration, persistent project context, branding integration, and the first usable GUI administration shell.

Phase Goal

Deliver a runnable, branded local application foundation that is secure enough to continue building on and structured enough to support Phase 2 Coinbase integration safely.

Purpose

Phase 1 converts the approved Phase 0 decisions into an actual local product baseline. The output of this phase should be a clean implementation starting point that preserves the approved constraints, keeps secrets out of source control, and gives the operator a first GUI shell for status, configuration, and future controls.

Carry-Forward Decisions

- Coinbase Advanced Trade remains the sole planned v1 exchange surface.
- Spot only and trade-only behavior remain mandatory for v1.
- No bank transfers, external wallet transfers, staking, Web3, margin, or derivatives may appear in the product scope.
- Live trading remains restricted to a screened subset of assets, not all Coinbase-listed assets by default.
- The current approved initial candidate class is BTC, ETH, SOL, XRP, ADA, LINK, LTC, XLM, BCH, and UNI, with DOGE on the conditional watchlist.
- Blockchain-derived data may be researched later as a signal source, but MEV-style or mempool-priority execution tactics remain out of scope for Coinbase Advanced spot.
- Bring-your-own-credentials and zero-secrets-in-repo remain required design rules.

Recommended GUI Stack

Phase 1 should use **Avalonia UI on .NET 8** as the recommended GUI stack.

- Avalonia is a better fit than Windows-only desktop options because Ubuntu is the primary runtime target and Windows 11 compatibility is still required.
- It keeps the product in the preferred .NET ecosystem without forcing a browser-hosted admin app for the first implementation phase.
- Phase 1 should treat this stack decision as part of the product baseline unless a later approved technical decision explicitly changes it.

Background

Phase 0 concluded that VectorQuay is technically feasible but should proceed with narrower scope because the current production account is a fragile low-capital starting point with confirmed Intro 1 Coinbase Advanced fees. Phase 1 therefore emphasizes product foundation, safety, and operator clarity rather than premature trading behavior.

In Scope

- Create the private GitHub repository for VectorQuay and establish it as the canonical source-control home for the project.
- Create the core `VectorQuay` repository and project structure for source, tests, docs, assets, and configuration templates.
- Establish the primary .NET solution baseline for Ubuntu-first local operation with a Windows 11 compatibility path.
- Implement the first GUI application shell for operator/admin use.
- Integrate branding, titling, and owner attribution into the application shell and documentation structure.
- Implement secure configuration loading and secret-handling patterns suitable for local development and future resale.
- Establish durable project context files and decision logging so future sessions can recover project state quickly.
- Define the first operator views and navigation structure for future Coinbase and strategy functionality.
- Establish a baseline local run experience and developer onboarding path.

Out of Scope

- Live Coinbase order placement or order cancellation.
- Paper trading.
- Strategy logic or portfolio optimization.
- AI-driven decisioning or sentiment analysis.
- Automated asset discovery jobs, except for reserving the shell and controls that later phases will use.
- Market-data persistence beyond what is needed for app-shell scaffolding.
- Production deployment automation.

Primary Users and Stakeholders

- Primary operator and owner: Brad Malia.
- Future local operator of the GUI administration application.
- Future resale customers who will need a reusable, credential-isolated installation path.
- Future engineering users who will build Coinbase integration, risk controls, and strategy components on top of this baseline.

Phase 1 Product Outcomes

- A structured local codebase exists and can be built and run consistently.
- A GUI shell exists with branded navigation and placeholder operator views.
- Secrets and environment-specific values are loaded from safe local sources rather than committed files.
- Project context, roadmap state, and decision logs are preserved in-repo in a repeatable way.
- The product foundation is ready for Phase 2 Coinbase read-only integration without reworking core layout or secret handling.

Functional Requirements

Repository and Solution Foundation

- Create the private GitHub repository in Phase 1 and commit the approved Phase 0 documentation baseline into it as the initial project record.
- Create a coherent root structure for source, tests, docs, assets, and configuration templates.
- Create the first .NET solution and projects needed for the GUI shell and shared application foundation.
- Preserve naming and structure that can grow into a resale-ready product rather than a single-user script repository.
- Include checked-in templates for configuration, but do not commit real credentials or machine-specific secrets.

GUI Administration Shell

- Provide a GUI application that runs locally on Ubuntu with a Windows 11 compatibility path, using Avalonia UI on .NET 8 as the default Phase 1 stack.
- Display VectorQuay branding and owner attribution appropriately.
- Provide a basic application shell with top-level navigation or sections for future views such as status, configuration, account, assets, decisions, and about/help.
- Support a safe startup state even before Coinbase connectivity exists.
- Clearly indicate that trading is not active during Phase 1.
- Include editable policy and configuration views, not just read-only placeholders, while still preventing secret values from being exposed in plain text once entered.

Configuration and Secret Handling

- Support local environment-specific configuration without storing secrets in the repository.
- Provide a clear pattern for loading Coinbase and OpenAI credentials later without baking them into code or docs.
- Support checked-in configuration templates and ignored local override files.
- Establish the approved local secret-storage path for Phase 1, including an ignored local configuration file and/or environment-variable pattern, rather than leaving secret storage unspecified.
- Define deterministic configuration precedence so later phases do not guess where effective values come from.
- For Phase 1, non-secret operator settings and policy edits should persist through application-managed local settings, checked-in files should remain templates only, and secrets should remain external to source control.
- Ensure application logging and error display do not expose secret values.
- Non-editable fields in the GUI should use an obvious disabled or greyed-out visual treatment so operators can distinguish fixed values from editable settings.

Context and Documentation Preservation

- Maintain the project context structure approved in Phase 0.
- Ensure major product, architecture, and implementation decisions can be captured durably for later sessions.
- Provide a clear place for current status, next steps, and handoff notes.

Operator Visibility

- Provide placeholder or scaffolded views for future operator needs, including status, allowed assets, protected assets, and application health.
- Make it clear which areas are informational in Phase 1 versus functional in later phases.
- Prepare the interface so future kill switches, connection health, and account views can be added without redesigning the shell.
- Reserve clear operator-facing sections for future holdings, investment allocation, trade history, KPIs, and charts so later phases can populate them without restructuring the application shell.
- Reserve UI structure for future trade-history filtering by timestamp, asset, action, and status, plus timestamp sorting.
- Reserve UI structure for future performance time-range selection such as 24H, 7D, 30D, 90D, 1Y, and ALL.
- Reserve UI structure for a combined activity view that can show historical decision records, decision rationale, and related execution outcomes in one place without forcing the operator to switch between separate trade and trace screens.
- Reserve UI structure for a future Risk and Thresholds view that separates operator-safe editable controls from advanced or read-only threshold settings and supports preset risk profiles such as High Risk, Medium Risk, Low Risk, and Custom.
- Reserve UI structure for future source-intelligence management, including a source registry that treats direct sources and watchers as first-class entries, filter controls, row selection, detail panel, ignore controls, reversible block controls, reactivate/unblock actions, reset-to-default actions, source reliability review, editable source scope, and source-state handling such as Active, Observed, Ignored, Blocked, and Needs Review.
- Reserve UI structure for future source discovery and onboarding, including operator-added sources by URL or handle plus AI-assisted watcher creation from prompts such as monitoring a subreddit, topic, or news theme, with those watchers appearing in the same registry as direct sources.
- Reserve a global alert and notification area in the shell plus a dedicated alerts view for future in-app alerts, notification history, and outbound alert preferences such as email and SMS.
- Reserve UI structure for alert preference granularity so operators can choose which alert categories, severities, channels, and delivery windows they want rather than receiving all alerts all of the time.

- Reserve UI structure for future update availability checks against a configured release or update source, including current version, latest available version, and user-facing update actions.
- Reserve UI structure for future asset-universe discovery that supports both scheduled automatic discovery runs and an operator-triggered manual refresh action.
- Reserve UI structure for future activity filtering by timestamp/date range, asset, action, source, status, and decision outcome.
- Allow standard trading metrics such as PnL, Win Rate, and Drawdown to retain their normal names while providing tooltip or inline-help explanations for operator clarity.
- Allow the operator to edit policy state that is safe to edit in Phase 1, including approved candidate assets, watchlist notes, protected-asset modes, and non-secret local configuration values.
- Reserve UI structure for policy selection and editing that makes the selected rule obvious, including checkbox-based selection or equivalent row-state clarity for future single-item and multi-item actions.
- Reserve UI structure for policy filtering so operators can narrow rules by asset, mode, state, or search term as the rule set grows.
- Non-secret policy edits in Phase 1 should be treated as authoritative local operator policy and persisted through application-managed local settings rather than editable checked-in files.
- Make clear that GUI edits do not bypass formal asset-governance rules or later technical enforcement boundaries.
- Switching from a Custom risk profile to a preset profile should require confirmation when unsaved custom changes would be overwritten.
- Reset to Default in threshold or source-management screens should explicitly restore the default recommended state rather than silently deleting context.

Non-Functional Requirements

- The application must be runnable locally on Ubuntu and structured for Windows 11 compatibility.
- Docker is optional and not required for Phase 1. The initial build should prioritize a straightforward native local development experience over containerization.
- Phase 1 must not require Docker to build, run, or configure the local application.
- The project must remain private-repo friendly and future resale friendly.
- No real secrets may be stored in source control, screenshots, sample config, or logs.
- The GUI shell should be usable and visually consistent with the approved branding direction.
- The codebase should be structured for maintainability and future testability rather than shortcut prototype behavior.
- A standard documented build-and-run workflow must exist for the primary development environment.
- Documentation updates must remain part of phase completion criteria.

Initial GUI Views

- **Home / Status:** operator dashboard view combining overall app state, safe inactive status, alerts, recent activity, top portfolio indicators, and top candidate visibility.
- **Configuration:** environment summary, editable non-secret configuration, secret-source status, and validation/error feedback without revealing secret values.
- **Assets:** an `Asset Universe` view showing the approved initial candidate class, DOGE watchlist note, future dynamic-universe direction, filtering by asset and state, default ranking by trade-priority score, reserved support for automatic discovery, and a manual refresh/discover action placeholder.
- **Policies:** trade-only rules, prohibited actions, and editable protected-asset settings with clear operator-facing terminology plus explicit row selection, filtering, and future bulk actions.
- **Portfolio:** reserved shell view for future holdings, allocation, and current investment state.
- **Activity:** reserved shell view for future decision-led activity records in one place, combining decision rationale with related execution outcomes and filtering by timestamp, asset, action, status, source, and outcome.
- **Performance:** reserved shell view for future KPIs, PnL summaries, graphs/charts, and selectable time ranges such as 24H, 7D, 30D, 90D, 1Y, and ALL.
- **Risk & Thresholds:** reserved shell view for future editable operator-safe risk controls, visible advanced thresholds, reset-to-default behavior, threshold explanations, and preset modes such as High Risk, Medium Risk, Low Risk, and Custom.
- **Sources:** reserved shell view for future source intelligence, including registry filters, checkbox-based row selection, source states, reliability review, weight overrides, ignore controls, reversible block/reactivate actions, reset-to-default actions that undo operator overrides, source details, editable scope, manual source addition, AI-assisted watcher creation, and watchers displayed as first-class registry items alongside direct sources.
- **Alerts:** reserved shell view for future in-app alerts, notification history, and outbound notification preferences such as email and SMS, including per-alert-type controls, severity filtering, and delivery scheduling or quiet-hour preferences.
- **About:** branding, owner attribution, version, documentation references, owner-configurable author-support options, and future update-management controls such as `Check for Updates` and `Download Update`.

Mock Screens

Mock Interaction Guidance

Phase 1 mock buttons are directional product controls, not final UI specifications. The PRD should define the intended behavior of meaningful operator actions, especially where misuse could affect policy, source handling, alerts, or future trading behavior. Final control flows, confirmations, and technical implementation details should be refined in the Phase 1 technical requirements document.

Home / Status Mock

VectorQuay

- Home
- Configuration
- Assets
- Policies
- Portfolio
- Activity
- Performance
- Risk & Thresholds
- Sources
- Alerts
- About

Global Alerts Area: reserved for future warnings, errors, and im
operator notifications.

VERSION 0.1.0

Dashboard

Tradi

App Health
Ready

Coinbase
Not Connected

Applicat
Pre-Integ

Portfolio Snapshot Last Updated: Placeholder

Total Value
Example Value

Open Positions
Placeholder

USD Available
Placeholder

T C

B A H

E A H

X A M P

Recent Activity

Config validated successfully · Placeholder

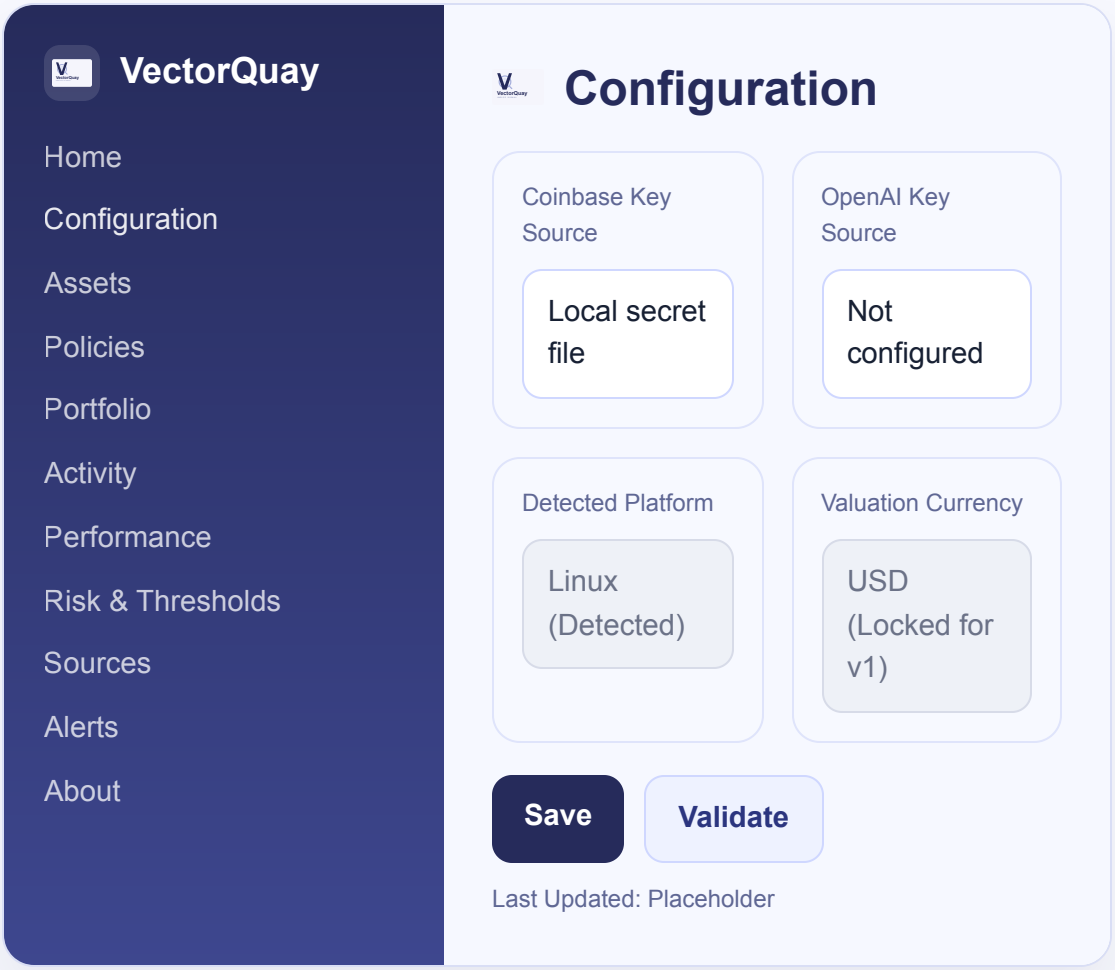
Asset universe reviewed · Placeholder

No live trades in Phase 1 · Placeholder

System Summary Last U Place

Phase 1 local shel No live trading, no trading, no exchar enabled. Next exp milestone: secure Coinbase read-on integration founda

Configuration Mock



Assets Mock

VectorQuay

- Home
- Configuration
- Assets
- Policies
- Portfolio
- Activity
- Performance
- Risk & Thresholds
- Sources
- Alerts
- About

Asset Universe

Filters: Asset / State

Default Sort: Trade Priority

Last Discovery Scan: Placeholder

Refresh Universe

Asset	State	Priority	Notes
BTC	Approved	High	USD preferred
ETH	Approved	High	Initial holding
XRP	Approved	Medium	Expansion set
DOGE	Watchlist	Low	Speculative
New Listing	Observed	Pending	Auto-discovered, not tradable yet

Newly discovered assets should appear automatically in a non-tradable state until screened. Operators should also be able to trigger a manual discovery refresh.

Last Screening Update: Placeholder

Policies Mock

VectorQuay

Home
Configuration
Assets
Policies
Portfolio
Activity
Performance
Risk & Thresholds
Sources
Alerts
About

Policies

Filters: Search / Asset / Mode / State

Default Sort: Most Recently Updated

Select one rule to edit details, or select multiple rules for future bulk actions.

Select	Asset	Mode	Notes
☑	ETH	Do Not Sell	Protected current holding
☐	BTC	Do Not Trade	Operator lock

Last Policy Update: Placeholder

Add Rule

Edit

Selected Policy Details

Asset: ETH

Mode: Do Not Sell

Notes: Protected current holding

Source: Operator rule

Last Updated: Placeholder

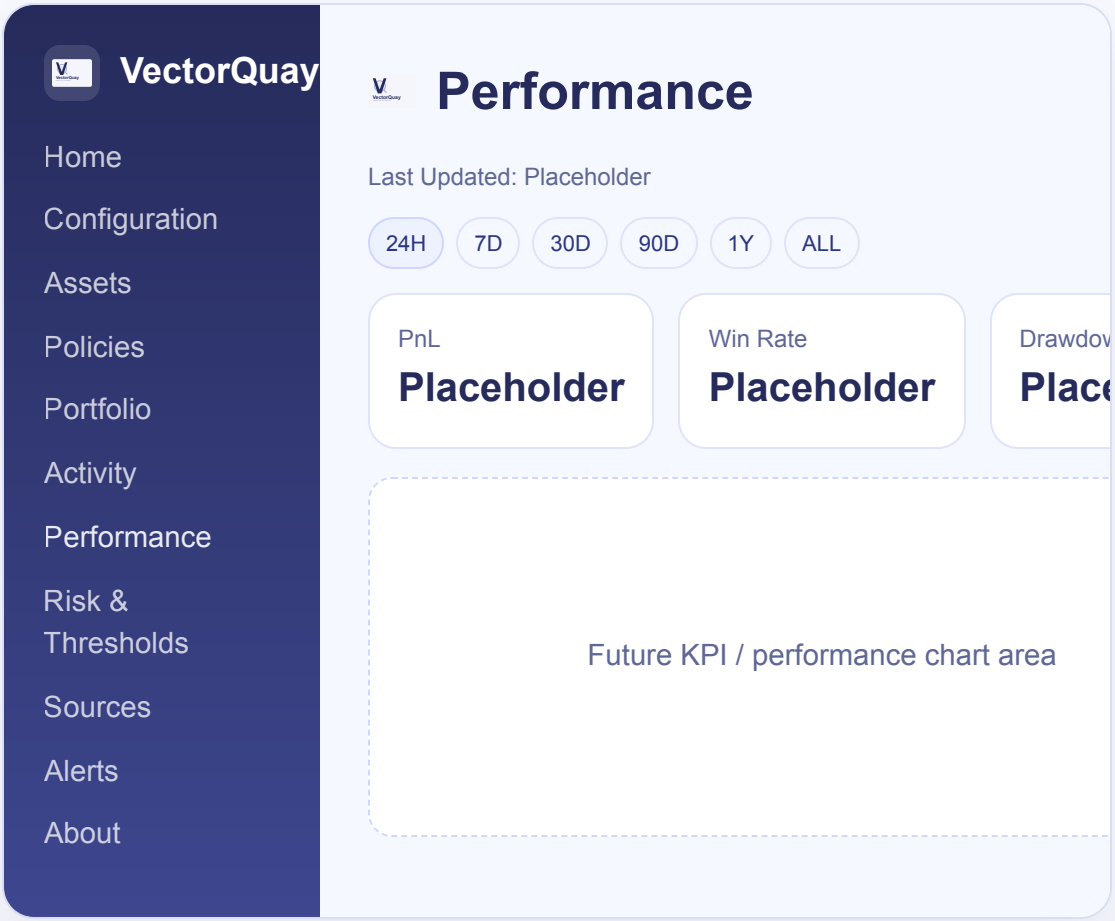
Edit Rule

Remove

Reset Default

Portfolio Mock

Performance Mock



Risk & Thresholds Mock

VectorQuay

- Home
- Configuration
- Assets
- Policies
- Portfolio
- Activity
- Performance
- Risk & Thresholds
- Sources
- Alerts
- About

Risk & Thresholds

Last Updated: Placeholder

High Risk | **Medium Risk** | Low Risk | Custom

Selected Profile: Medium Risk

Preset profiles should be recommended defaults. High, Medium, and Low should load suggested values. Custom should allow operators to edit the values directly.

Operator-Safe Controls

Max Daily Loss · Preset Value

Max Position Size · Preset Value

Max Portfolio Exposure · Preset Value

Advanced Thresholds

Spread Tolerance · Visible / Read-Only Placeholder

Liquidity Minimum · Visible / Read-Only Placeholder

Confidence Cutoff · Visible / Read-Only Placeholder

Save | **Switch to** | **Reset to**

Custom

Default

Activity Mock

VectorQuay

Home

Configuration

Assets

Policies

Portfolio

Activity

Performance

Risk & Thresholds

Sources

Alerts

About

Activity

Filters: Date Range / Asset / Action / Source / Outcome

Sort: Timestam First)

Timestamp	Asset / Pair	Action	Amount	Outcome
Placeholder	BTC-USD	Buy	Placeholder	Selected
Placeholder	ETH-USD	Sell	Placeholder	Rejected

Selected row shown below. Viewing Decision: BTC-USD Buy · Amoun Timestamp Placeholder

Decision Details

Reserved for future rationale summary, confidence scor pass/fail checks from the strategy and risk engine.

Policy and Risk Checks

Reserved for future permissions, protected-asset checks, sizing che gate results tied to this exact decision.

Source	Signal	Weig
Market Trend	Positive	Defau
Influencer A	Negative	Overr
News Feed B	Positive	Defau

Sources Mock

VectorQuay

- Home
- Configuration
- Assets
- Policies
- Portfolio
- Activity
- Performance
- Risk & Thresholds
- Sources
- Alerts
- About

Sources

Last Updated: Placeholder

Active Sources
Placeholder

Needs Review
Placeholder

All | Direct Sources | Watchers

Filters: Search / Type / State / Asset / More Filters

Default Sort: Most Influential

Add Source | Add Watcher

Select	Source	Type	State	Weight
☑	Influencer A	Influencer	Active	Reduced Override
☐	News Feed B	News Site	Active	Default
☐	/r/dogecoin	Watcher	Observed	Pending
☐	Influencer C	Influencer	Ignored	0

Click a row or checkbox to select a source or watcher. Use one selection for detail editing, or select multiple rows for future bulk actions.

Selected: Influencer A

Identity

Type: Influencer

State: Active

Scope

DOGE, SOL

Influence

Weight: Reduced Override

Last Contribution:

Placeholder

Edit Weight | Edit Scope | More Actions

More Actions: Ignore / Block / Reactivate / Reset Default

View History | Automation Settings

History should show lifecycle events, override changes, and recent contributions. AI-created watchers should begin as Observed or Needs Review. State actions should remain reversible.

Alerts Mock

VectorQuay

- Home
- Configuration
- Assets
- Policies
- Portfolio
- Activity
- Performance
- Risk & Thresholds
- Sources
- Alerts
- About

Alerts

Recent Alerts

New Version Available · Placeholder

Large Loss Threshold Triggered · Placeholder

System Validation Completed · Placeholder

Large Win Threshold Triggered · Placeholder

Notification Preferences

In-App Alerts: Enabled

Email Alerts: Enabled

SMS Alerts: Disabled

Enabled Types: Loss, System, Version

Quiet Hours: 10:00 PM - 7:00 AM

Edit Preferences

Send Test

Edit Preferences should let the operator choose alert types, severity levels, delivery channels, and time windows so notifications stay useful rather than noisy.

About Mock

VectorQuay

- Home
- Configuration
- Assets
- Policies
- Portfolio
- Activity
- Performance
- Risk & Thresholds
- Sources
- Alerts
- About

VectorQuay

Indigo Signal · Author/Owner: Brad Malia

About This Application

Phase 1 local administration shell for the VectorQuay Coinbase spot trading platform.

Version: 0.1.0 placeholder · Trading Status: Inactive · Documentation: Roadmap, PRD, TRD, and test plans linked here in later implementation.

Updates

Current Version: 0.1.0 · Latest Available: Placeholder

Check for Updates

Download Update **View Release**

Phase 1 reserves the update flow in the shell. Safe version checks and download/open behavior should work against a configured release source before any in-app self-update behavior is considered.

Support the Author

QR Code

Manual crypto tip only. Not connected to bot trading behavior. This section should be owner-configurable and may be disabled or omitted in resale-oriented builds.

Wallet Address Placeholder

Copy Address **Supported Assets**

Success Metrics

- The codebase can be cloned, configured with templates, and launched locally without ad hoc setup steps.
- The private GitHub repository exists and contains the approved documentation baseline plus the initial Phase 1 project scaffold.
- The GUI shell builds and runs on the primary target environment.
- Secrets are externalized and redacted appropriately in logs and UI.
- Configuration precedence and secret-source behavior are documented and deterministic.
- Editable policy and configuration screens behave coherently and communicate invalid or incomplete states clearly.
- Missing-secret or not-yet-configured states are handled gracefully and visibly in the UI.
- The documentation structure remains synchronized with the codebase structure.
- Phase 2 can begin Coinbase read-only integration without foundational restructuring.

Deliverables

- Approved Phase 1 PRD.
- Phase 1 technical requirements document.
- Phase 1 unit, integration, and regression test plans.
- Private GitHub repository initialized with the approved documentation baseline.
- Initial repository and solution scaffold.
- Runnable GUI administration shell.
- Secure configuration and secret-handling baseline.
- Updated project context and decision-log artifacts.

Acceptance Criteria

Phase 1 Is Complete When

- The repository contains a coherent .NET application structure and supporting docs structure.
- The private GitHub repository exists and contains the approved Phase 0 baseline plus the Phase 1 foundation work.
- The GUI shell launches locally and shows the initial operator/admin sections.
- No secrets are present in the repository and configuration templates are in place.
- The Phase 1 GUI stack decision is explicit and reflected in the implementation baseline.
- Editable policy and configuration surfaces exist for safe local settings and non-secret operator controls.
- Ignored local configuration paths are established so secret-bearing files cannot be committed accidentally through normal workflow.
- The PRD clearly distinguishes authoritative local policy state from informational UI state.
- The approved Phase 0 constraints are visible in the product foundation rather than only in documents.
- The product is ready for safe Phase 2 Coinbase read-only integration work.

Risks

Phase 1 Risks

- Choosing a GUI approach that works poorly across Ubuntu and Windows can create avoidable rework.
- Weak secret handling in Phase 1 would contaminate later phases and future resale readiness.
- An under-structured codebase would force later refactoring before Coinbase integration and testing.
- Overbuilding the GUI shell now could delay more important Phase 2 integration work.

Recommended Next Step

Proceed to the Phase 1 technical requirements document. That document should convert this approved product intent into concrete architecture, project structure, GUI technology, secret handling, and implementation/test-planning guidance.

VectorQuay Phase 1 PRD • In review • Author/Owner: Brad Malia